Counterexamples to Five Conjectured Upper Bounds for the Largest Laplacian Eigenvalue

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Abstract

Brankov, Hansen, and Stevanović (2006) used the AutoGraphiX system to generate 68 conjectured upper bounds for the largest Laplacian eigenvalue $\mu(G)$ of a graph G. Ghebleh et al. (2024) disproved 30 of these, leaving 38 open. We resolve five of the remaining open conjectures (Bounds 11, 13, 40, 45, and 48) by constructing explicit counterexample families—star-of-cliques, double star, and caterpillar graphs—whose Laplacian spectral radii we compute analytically via equitable partition quotient matrices. We identify hub–periphery degree imbalance as the structural mechanism underlying all violations and classify the remaining 33 bounds by their resistance to this mechanism.

Keywords: Laplacian eigenvalue, spectral radius, upper bound, counterexample, equitable partition
2020 MSC: 05C50, 15A18

1. Introduction

Let $G = (V, E)$ be a simple connected graph on $n = |V|$ vertices and $m = |E|$ edges. The *Laplacian matrix* of G is $L(G) = D(G) - A(G)$, where $A(G)$ is the adjacency matrix and $D(G) = \operatorname{diag}(d_1, \ldots, d_n)$ is the diagonal matrix of vertex degrees. Since $L(G)$ is positive semidefinite with row sums zero, its eigenvalues satisfy $0 = \lambda_n \leq \cdots \leq \lambda_1$, and the largest eigenvalue

$$\mu(G) \coloneqq \lambda_{\max}\big(L(G)\big) \tag{1}$$

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is called the *Laplacian spectral radius*. Upper bounds on $\mu(G)$ in terms of easily computed graph invariants are of both theoretical and algorithmic interest; see [1] for background. In 2006, Brankov, Hansen, and Stevanović [2] used the AutoGraphiX system [3] to generate 68 conjectured upper bounds for $\mu(G)$ expressed solely in terms of vertex degrees d_v and *average neighbor degrees*

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u. \tag{2}$$

The 68 conjectures fall into two families. *Vertex-type* bounds have the form $\mu(G) \leq \max_v f(d_v, m_v)$, and *edge-type* bounds have the form

$$\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j).$$

None had been proved or disproved for nearly two decades.

Ghebleh, Al-Yakoob, Kanso, and Stevanović [4] recently applied the cross-entropy method (CEM) of Wagner [5] to disprove 30 of the 68 bounds by constructing small counterexample graphs, leaving 38 bounds open. Their counterexamples all have at most 13 vertices and were found by training a neural network to generate adjacency matrices that violate a given bound. Subsequently, Bouffard and Breen [6] parallelized the CEM approach and extended the search to larger graphs, but reported no new counterexamples among the remaining 38 bounds. Roucairol and Cazenave [7] applied nested Monte Carlo search to the same problem, likewise without resolving any of the open cases.

In the present paper we disprove five of the 38 remaining open bounds.

Theorem 1.1 (Main result)**.** *BHS Bounds 11, 13, 40, 45, and 48 from [2] are false. Specifically:*

(a) *Bound 11 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *and* $t \geq 2m - 1$ (Theorem 4.2).

(b) *Bound 13 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and sufficiently large* t (Theorem 4.3).

(c) *Bound 40 is violated by the star of stars* $\mathrm{SoS}(39, 8)$ *and* $\mathrm{SoS}(49, 9)$ (Theorem 4.7).

(d) *Bound 45 is violated by* $\mathrm{DS}(k, k)$ *for all* $k \geq 6$ (Theorem 4.5).

(e) *Bound 48 is violated by the path* P_3 (Remark 4.8).

In each case the proof is constructive: we exhibit an infinite family (or, for Bound 48, a single small graph) together with an analytic computation of its Laplacian spectral radius via equitable partition quotient matrices.

Our approach differs fundamentally from the optimization-based methods of [4, 5, 6]. Rather than training a neural network to search for counterexamples, we analyze the algebraic structure of each bound formula and identify graph topologies that exploit a specific weakness. The key observation is that every bound we disprove exhibits high sensitivity to the ratio m_v/d_v at vertices (or pairs of adjacent vertices) where these quantities diverge. Graphs with *hub–periphery* structure—a high-degree hub connected to dense peripheral subgraphs—create exactly this divergence: the hub's average neighbor degree m_v is determined by the peripheral structure and can be made large relative to the hub degree d_v, or vice versa. Previous computational searches missed these counterexamples because CEM and exhaustive enumeration up to $n = 13$ vertices tend to explore near-regular graphs, where $m_v \approx d_v$ for all v and the bounds are naturally tight. The smallest counterexample we find for Bound 11 has $n = 29$ vertices and the largest has $n = 172$, well beyond the reach of exhaustive search.

The paper is organized as follows. Section 2 fixes notation and recalls the theory of equitable partitions. It also states the five BHS bounds that we disprove. Section 3 defines the counterexample graph families: star of cliques, double star, and star of stars. Section 4 contains the main results: closed-form spectral radius computations and the violation proofs for each of the five bounds. Section 5 analyzes the hub–periphery mechanism common to all five violations. Section 6 classifies the remaining 33 open bounds by their resistance to this mechanism. Section 7 discusses implications for automated conjecture generation and directions for future work.

2. Preliminaries

2.1. Graph notation

Throughout this paper, $G = (V, E)$ denotes a simple connected graph with vertex set V, edge set E, $n = |V|$ vertices, and $m = |E|$ edges. The *adjacency matrix* $A(G)$ is the $n \times n$ matrix with $A_{ij} = 1$ if $\{i, j\} \in E$ and $A_{ij} = 0$ otherwise. The *degree matrix* is $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_v = \sum_u A_{vu}$ is the degree of vertex v. The *Laplacian matrix* $L(G) =$

$D(G) - A(G)$ is positive semidefinite with smallest eigenvalue 0. We write $\mu(G) = \lambda_{\max}(L(G))$ for the *Laplacian spectral radius*.

For each vertex $v \in V$ with $d_v \geq 1$, the *average neighbor degree* is

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u,$$

where the sum is over all neighbors u of v. Note that $m_v \geq 1$ for every vertex in a connected graph, and $m_v = d_v$ for every vertex if and only if G is regular.

2.2. Equitable partitions

Equitable partitions provide a powerful tool for computing eigenvalues of highly structured graphs by reducing the $n \times n$ Laplacian to a small quotient matrix. We recall the essential definitions; for a comprehensive treatment see Godsil and Royle [8].

Definition 2.1 (Equitable partition)**.** A partition $\pi = \{C_1, \ldots, C_k\}$ of the vertex set $V(G)$ is *equitable* if, for every pair (i, j) with $1 \leq i, j \leq k$, every vertex in C_i has the same number of neighbors in C_j. We denote this common value by b_{ij}.

Given an equitable partition with cells $C_1, \ldots, C_k$, the *quotient matrix* is the $k \times k$ matrix Q with entries $Q_{ij} = b_{ij}$, and the *quotient Laplacian* is the $k \times k$ matrix Q_L with entries

$$(Q_L)_{ij} = \begin{cases} d_{C_i} & \text{if } i = j, \\ -b_{ij} & \text{if } i \neq j, \end{cases}$$

where $d_{C_i} = \sum_{j=1}^{k} b_{ij}$ is the common degree of every vertex in cell C_i.

Theorem 2.2 ([8, Theorem 9.3.3])**.** *Let π be an equitable partition of G with quotient Laplacian Q_L. Then every eigenvalue of Q_L is an eigenvalue of $L(G)$. In particular,*

$$\mu(G) \geq \lambda_{\max}(Q_L),$$

with equality whenever the eigenvector of $L(G)$ corresponding to $\mu(G)$ is constant on each cell of π.

In all counterexample families considered in this paper, the natural partition by vertex type is equitable, and the largest eigenvalue eigenvector is piecewise constant on cells. This reduces the computation of $\mu(G)$ to finding the largest root of a polynomial of degree at most 3.

2.3. The BHS bounds

We now state the five conjectured bounds from [2] that we disprove. All five are listed as open in [4, Table 1]. Bounds 11 and 13 are vertex-type bounds; Bounds 40, 45, and 48 are edge-type bounds.

Conjecture 2.3 (BHS Bound 11 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^3}{d_v^2}.$$

Conjecture 2.4 (BHS Bound 13 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^4}{d_v^3}.$$

Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j) \in E} \left(2 + \sqrt{2(m_i - 1)^2 + 2(m_j - 1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

Conjecture 2.6 (BHS Bound 45 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j) \in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right).$$

Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j) \in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

Bounds 11 and 13 are structurally similar: both take the form $\max_v 2m_v^p / d_v^{p-1}$ with $p = 3$ and $p = 4$, respectively. When m_v/d_v is large, these bounds grow as a power of the ratio m_v/d_v, making them vulnerable to graphs where some vertex has high average neighbor degree relative to its own degree. Bounds 40 and 45 are edge-type bounds with more complex dependence on both d_v and m_v; their vulnerability is subtler and requires careful asymptotic analysis. Bound 48 is unusual in that it is violated by the path P_3, the smallest non-trivial connected graph; we conjecture that this bound was generated under a minimum order assumption that excluded $n = 3$.

3. Counterexample graph families

Definition 3.1 (Star-of-Cliques)**.** For integers $m \geq 3$ and $t \geq 2$, the graph $\mathrm{SoC}(K_m, t)$ consists of a central *hub* vertex v_0 connected by a single edge to one designated vertex (the *connector*) in each of t disjoint copies of the complete graph K_m. The total number of vertices is $n = 1 + tm$.

Definition 3.2 (Double Star)**.** For an integer $k \geq 1$, the double star $\mathrm{DS}(k, k)$ is obtained by connecting two vertices u and v, each of which is adjacent to k pendant vertices. The total number of vertices is $n = 2k + 2$.

Definition 3.3 (Star-of-Stars)**.** For integers $t \geq 2$ and $s \geq 1$, the graph $\mathrm{SoS}(t, s)$ consists of a central *hub* vertex v_0 adjacent to t *intermediate* vertices $w_1, \ldots, w_t$, each of which is adjacent to s pendant *leaf* vertices. The total number of vertices is $n = 1 + t + ts = 1 + t(1 + s)$.

Definition 3.4 (Caterpillar)**.** For integers $\ell \geq 2$ and $k \geq 0$, the caterpillar $\mathrm{Cat}(\ell, k)$ is obtained from a path on ℓ vertices (the *spine*) by attaching k pendant leaves to each spine vertex. The total number of vertices is $n = \ell(k + 1)$.

4. Main results

Proposition 4.1. *For $m \geq 3$ and $t \geq 2$, the Laplacian spectral radius of $\mathrm{SoC}(K_m, t)$ is*

$$\mu\big(\mathrm{SoC}(K_m, t)\big) = \frac{(m + t + 1) + \sqrt{(t - m)^2 + 2(t + m) - 3}}{2}. \tag{3}$$

Proof. The vertex set admits an equitable partition into three cells: $H = \{v_0\}$ (hub), C (connectors, $|C| = t$), and I (interior, $|I| = t(m-1)$). The quotient Laplacian is the 3×3 matrix

$$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix}. \tag{4}$$

One eigenvalue is $\lambda_1 = 0$ (since the quotient Laplacian has zero row sums). The remaining two eigenvalues are roots of

$$\lambda^2 - (m + t + 1)\lambda + (tm + 1) = 0,$$

giving the largest eigenvalue as (3). Since the corresponding eigenvector is piecewise constant on the cells, this is also the largest eigenvalue of $L(G)$. $\square$

Theorem 4.2. *BHS Bound 11,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2}, \tag{5}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *and* $t \geq 2m - 1$. *In particular,* $\mathrm{SoC}(K_8, 19)$ *with* $n = 153$ *satisfies* $\mu(G) \approx 20.557$ *while the bound gives* ≈ 19.191.

Proof. By Proposition 4.1, we have $\mu = \frac{(m+t+1)+\sqrt{(t-m)^2+2(t+m)-3}}{2}$. The hub vertex v_0 has $d_{v_0} = t$ and $m_{v_0} = m$ (all neighbors are connectors of degree m), so Bound 11 evaluates to

$$\frac{2m_{v_0}^3}{d_{v_0}^2} = \frac{2m^3}{t^2}.$$

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2 + 4m - 4}\big)/2 > 2m$ since $\sqrt{m^2 + 4m - 4} > m$ for $m \geq 2$ (obtained by squaring). For $t > 2m - 1$ the connector bound $2m_v^3/m^2$ grows while μ also grows, and the gap widens. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m - 1$. *In particular,* $\mathrm{SoC}(K_9, 19)$ *with* $n = 172$ *satisfies* $\mu(G) \approx 20.685$ *while the bound gives* ≈ 19.845.

Proof. The connector vertex has $d = m$ and $m_v = \big(t + (m-1)^2\big)/m$. At $t = 2m - 1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m+\sqrt{m^2 + 4m - 4})/2$. Since $\sqrt{m^2 + 4m - 4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$, the interior bound is ≈ 10.195 but $\mu\big(\mathrm{SoC}(K_5, 9)\big) \approx 10.702 > 10.195$. □

Proposition 4.4. *For* $k \geq 2$, *the Laplacian spectral radius of* $\mathrm{DS}(k, k)$ *is*

$$\mu\big(\mathrm{DS}(k, k)\big) = \frac{(k+3) + \sqrt{k^2 + 6k + 1}}{2}. \tag{7}$$

Proof. The $\mathbb{Z}_2$ symmetry swapping the two hubs decomposes eigenspaces into symmetric and antisymmetric sectors.

Symmetric sector. Quotient matrix $Q_+ = \left(\begin{smallmatrix} k & -k \\ -1 & 1 \end{smallmatrix}\right)$, eigenvalues 0 and $k+1$.

Antisymmetric sector. Quotient matrix $Q_- = \left(\begin{smallmatrix} k+2 & -k \\ -1 & 1 \end{smallmatrix}\right)$, characteristic polynomial $\lambda^2 - (k+3)\lambda + 2 = 0$, eigenvalues $(k+3 \pm \sqrt{k^2+6k+1})/2$.

Pendant leaves contribute eigenvalue 1 with multiplicity $2(k-1)$. Since $\sqrt{k^2+6k+1} > k-1$ for $k \geq 2$, the antisymmetric λ_+ is the global maximum. □

Theorem 4.5. *BHS Bound 45,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right), \quad (8)$$

is violated by $\mathrm{DS}(k,k)$ *for all* $k \geq 6$. *At* $k = 6$, $\mathrm{DS}(6,6)$ *with* $n = 14$ *satisfies* $\mu(G) \approx 8.772$ *while the bound gives* ≈ 8.676.

Proof. The hub–leaf edge maximizes Bound 45 for $k \geq 6$. Showing $\mu > B_{45}^{\mathrm{hl}}$ and squaring twice reduces to $4k(k^3 + 8k^2 - 11k - 2) > 0$, which holds for $k \geq 2$ since $k^3 + 8k^2 - 11k - 2 > 0$ for $k \geq 2$. The gap satisfies $B_{45} - \mu = -1/(2k) + O(k^{-2})$. □

Proposition 4.6. *For* $t \geq 2$ *and* $s \geq 1$, *the Laplacian spectral radius of* $\mathrm{SoS}(t,s)$ *is*

$$\mu\big(\mathrm{SoS}(t,s)\big) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}. \quad (9)$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \left(\begin{smallmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{smallmatrix}\right)$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.7. *BHS Bound 40,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i - 1)^2 + 2(m_j - 1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right), \quad (10)$$

is violated by $\mathrm{SoS}(39,8)$ *with* $n = 352$, *where* $\mu(G) \approx 40.256$ *and the bound gives* ≈ 38.982.

Table 1: Summary of counterexamples to BHS bounds. "Gap" denotes the bound value minus $\mu(G)$ (negative = violation).

Bound	Counterexample	n	$\mu(G)$	Bound value	Gap
11	$\mathrm{SoC}(K_8, 19)$	153	20.5574	19.1914	−1.3660
13	$\mathrm{SoC}(K_9, 19)$	172	20.6847	19.8447	−0.8399
40	$\mathrm{SoS}(39, 8)$	352	40.2560	38.9818	−1.2741
45	$\mathrm{DS}(6, 6)$	14	8.7720	8.6762	−0.0958
48	P_3	3	3.0000	2.9289	−0.0711

Proof. The hub–intermediate edge has $d_h = 39$, $m_h = 9$, $d_w = 9$, $m_w = 47/9$. By Proposition 4.6, $\mu = (49 + \sqrt{993})/2$. The inequality $\mu > B_{40}$ reduces to $\sqrt{993} > 27.25$, i.e., $993 > 742.6$, which holds. □

Remark 4.8 (Bound 48 — degenerate case). BHS Bound 48,

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}},$$

is violated by P_3: the bound gives $10/(2+\sqrt{2}) \approx 2.929$ while $\mu(P_3) = 3$. This is the only counterexample found.

5. Why these bounds fail

5.1. Hub-invisible bounds

Every vertex-type BHS bound has the form $\mu(G) \leq \max_{v\in V} f(d_v, m_v)$ for some rational function f. On a star-of-cliques $\mathrm{SoC}(K_m, t)$ the maximum is taken over three vertex types (hub, connector, interior), and the bound value is the largest of the three contributions.

Definition 5.1 (Hub-invisible)**.** A bound $B(G) = \max_v f(d_v, m_v)$ is *hub-invisible* if $f(t, m) \to 0$ as $t \to \infty$ (m fixed), and *hub-visible* if $f(t, m) \to \infty$.

Proposition 5.2 (Hub-invisible bounds fail)**.** (a) ***Bounds 11, 13*** *are hub-invisible:* $f_{11} = 2m^3/t^2 \to 0$, $f_{13} = 2m^4/t^3 \to 0$.

(b) ***Bounds 14 and 16*** *are hub-visible:* $f_{14}^{\mathrm{hub}} = 2t^3/m^2 \to \infty$, $f_{16}^{\mathrm{hub}} = 2t^4/m^3 \to \infty$.

(c) ***Bound 30*** *is hub-visible:* $f_{30}^{\text{hub}} = m^3/t^2 + t^2/m \sim t^2/m \to \infty$.

Proof. Substitute $(d_v, m_v) = (t, m)$ into each formula and take $t \to \infty$. □

The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub—which dominates the spectrum—becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, while $\mu > 2m$ (Theorem 4.2).

Remark 5.3. Bound 40 fails on $\mathrm{SoS}(t, s)$ by a related mechanism: the hub–intermediate edge creates a transient violation window at $t \sim s^2$, closing for both small and large t.

5.2. Edge-based antisymmetric modes

Bound 45 fails on $\mathrm{DS}(k, k)$ because the spectral radius arises from the σ-antisymmetric sector (Proposition 4.4), where the two hubs carry opposite signs. No single edge evaluation in the bound formula captures this global oscillation. The gap $\mu - B_{45}$ is $O(1/k)$, suggesting repairability with an additive $O(1/\Delta)$ correction.

5.3. Characterizing vulnerable bounds

Conjecture 5.4. *A vertex-type BHS bound* $\mu(G) \leq \max_v f(d_v, m_v)$ *is violated by some hub–periphery graph if and only if:* (i) f *is monotonically increasing in* m_v/d_v*, and* (ii) f *is regularity-tight:* $f(k, k) = 2k$ *with no margin.*

5.4. A weighted Laplacian intermediate bound

In attempting to prove the tight bounds (33, 42, 47), we obtained the following intermediate result.

Proposition 5.5 (Weighted Laplacian bound)**.** *For any connected graph* G*,*

$$\mu(G)^2 \leq \lambda_{\max}(M),$$

where M *is the Laplacian of the weighted graph on the same vertex set with edge weights* $w_{ij} = d_i + d_j$ *for each* $\{i, j\} \in E$*.*

Table 2: Classification of the 33 remaining open BHS bounds.

Category	Count	Bounds	Assessment
SUPER-SAFE	8	1, 4, 5, 7, 14, 16, 21, 30	Likely provable
SAFE	20	6, 8, 9, 12, 18–20, 22, 24–27, 34, 35, 37–39, 44, 46, 56	Plausible
TIGHT	3	33, 42, 47	Sharp; provable?
MARGINAL	2	10, 23	Undetermined

Proof sketch. Let x be the eigenvector of $L(G)$ with eigenvalue $\mu(G)$, normalized so that $\|x\|_2 = 1$. The eigenequation gives $\mu(G)\, x_v = d_v x_v - \sum_{u\sim v} x_u$. Squaring both sides:

$$\mu(G)^2\, x_v^2 = \Big(d_v x_v - \sum_{u\sim v} x_u\Big)^2.$$

Applying Cauchy–Schwarz to $\big(\sum_{u\sim v} x_u\big)^2 \le d_v \sum_{u\sim v} x_u^2$ in the expanded square and summing over all vertices yields $\mu(G)^2 \le x^T M\, x$, where M has entries

$$M_{vv} = \sum_{u\sim v}(d_v + d_u), \qquad M_{vu} = -(d_v + d_u) \ \text{ if } u \sim v.$$

The Rayleigh quotient bound gives $x^T M\, x \le \lambda_{\max}(M)$. □

Remark 5.6. Proposition 5.5 implies that proving Bound 42, for example, reduces to showing $\lambda_{\max}(M) \le \max_{(i,j)\in E}(d_i^2 + d_j^2 + 2m_i \cdot m_j)$. This second inequality remains open.

6. Classification of the remaining 33 bounds

We classify the 33 remaining bounds by gap behavior on the tested families.

The SUPER-SAFE bounds include all hub-visible bounds (14, 16, 30). The three TIGHT bounds achieve exact equality on P_3 and complete bipartite graphs, making them the most promising candidates for a proof.

6.1. Exhaustive verification of the tight bounds

We performed exhaustive computational verification of the three tight bounds (33, 42, 47) on every connected graph with $n \le 10$ vertices. The

total number of connected graphs tested was 11,988,768 (11,117 for $n = 8$; 261,080 for $n = 9$; 11,716,571 for $n = 10$). Zero violations were found for any of the three bounds.

Theorem 6.1 (Finite verification)**.** *Bounds 33, 42, and 47 from [2] hold for all connected graphs with at most* 10 *vertices.*

Among all graphs tested, the tightest instances—those achieving exact equality in all three bounds simultaneously—are the complete bipartite graphs $K_{a,b}$. On regular bipartite graphs, the vertex and edge parameters satisfy $d_i = d_j$ and $m_i = m_j = d_i$, causing the bound formulas to collapse to $\mu(G) = 2\Delta$, which is the known spectral radius of $K_{a,a}$.

Windmill graphs $W(k, 3)$ (the friendship graphs) provide asymptotically tight instances for Bound 42: the relative gap satisfies $\big(B_{42}-\mu\big)/\mu \sim \sqrt{3}/k \to 0$ as $k \to \infty$.

7. Discussion

Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [4] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction, leaving 33 open.

Structural vs. computational approaches.. Cross-entropy search explores adjacency matrices for $n \leq 24$. Our smallest counterexample has $n = 29$, beyond exhaustive reach. Structural construction targets algebraic weaknesses (hub-invisible ratios, antisymmetric modes) and produces infinite families with closed-form spectral radii. The two approaches are complementary.

Gap asymptotics.. Bounds 11/13: gap diverges (irreparable). Bound 40: transient window (max gap 1.74). Bound 45: gap $\to 0$ as $1/(2k)$ (repairable). Bound 48: degenerate P_3 case only.

Proof attempts for tight bounds.. We attempted to prove the three tight bounds (33, 42, 47) using Rayleigh quotient methods, Schur complement analysis, Gershgorin circles, and degree sequence constraints. While a key intermediate result was obtained (Proposition 5.5), complete proofs remain elusive. The fundamental obstacle is that standard spectral techniques do not naturally produce edge-maximum bounds involving average neighbor degrees: the Rayleigh quotient $x^T Lx/x^T x$ yields vertex-sum expressions, whereas edge-type bounds require converting these into edge-maximum form.

Open questions.. (a) Can any of the 33 remaining bounds be proved, especially the tight bounds 33, 42, 47? (b) Do three-level hierarchies violate additional bounds? (c) Can Conjecture 5.4 be proved? (d) Can Bound 45 be repaired with an $O(1/\Delta)$ correction?

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Claude (Anthropic; model: Claude Opus 4, 2025) for computational assistance, including code development for graph construction and eigenvalue computation, structural analysis of graph families, and systematic counterexample search. The author reviewed and edited all content and takes full responsibility for the publication.

Acknowledgments

The author thanks Pierre Hansen and Dragan Stevanović for making the BHS bound list publicly available. Computational assistance was provided by Claude (Anthropic); all mathematical results were independently verified by the author.

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